

DEFINITIONS · AXIOMS · THEOREMS

Double-Slit Geometry

*A stand-alone essay on the geometry of cross-slit projection. Every figure below is **computed**, not sketched: the curves are generated from the calibrated rig by the same script that verifies the theorems (`check_geometry_essay.js`), so a diagram and its theorem cannot disagree. Companion to the formal document [\[8\]](#).*

WHAT A GEOMETRY IS

A geometry, in the sense this essay uses the word, is two things at once. In the synthetic tradition of Euclid and Hilbert, it is a set of primitives, a list of axioms about their incidence, and the theorems those axioms force. In the tradition of Klein's Erlangen program^[5], it is a space together with a group of transformations, and its theorems are the statements that survive the group. The two views rank the familiar geometries on one ladder: projective geometry is the geometry of the full collineation group, with the cross-ratio as its fundamental invariant; affine geometry fixes the plane at infinity and gains parallelism and ratio; Euclidean geometry fixes more and gains length.

Double-slit geometry takes its place in this family. It is what remains invariant when the only structure held sacred is **a pair of skew lines** — the two slits. The guiding fact, stated here once and proved below: a pinhole is a *point* through which all rays concur; a double slit is the exact refusal of that concurrence. No two generic rays of the system meet anywhere. Everything that follows is the systematic study of that refusal.

Honesty about provenance, before anything is claimed. The underlying objects are classical: the family of transversals to two skew lines is the hyperbolic linear congruence, studied since Plücker, Kummer, and Sturm^[4]; the camera model appears in the modern imaging literature as the crossed-slits projection of Zomet, Feldman, Peleg, and Weinshall^[1] and as a class in Yu and McMillan's General Linear Camera taxonomy^[2]; and the axiom system below instantiates the theory of chain geometries and Minkowski planes built by the school of

Walter Benz^{[3], [7]}. What this body of work contributes is the assembly — the presentation of the camera as a geometry in both of Klein's senses — and Theorem 7, which identifies a working codec's formula as a cross-ratio.

Illustration placeholder P1 — the physical fact. Insert the paired cardboard-box sketches (perspective experiment / cross-slit experiment) and the "basic idea" exploded diagram from *Cross-slit-projection.pdf*, p. 2. Hand drawing preferred here: the essay's one non-computed figure should be the one that shows the geometry began as cardboard.

II THE MODEL

Work in real projective 3-space P^3 — projective, so that "parallel" is not an exception but a location; the affine and metric layers are recovered in section X by nominating special elements. Fix two skew lines ℓ_1 and ℓ_2 , the **slits**. In the calibrated rig used for every computation in this essay, ℓ_1 is the vertical slit in the plane $z = Z_v = 150$, ℓ_2 the horizontal slit in the plane $z = Z_h = 250$, and the screen is the plane $z = 0$; in the Tetra coordinate system the slits are one pair of opposite edges of the root tetrahedron. Nothing below depends on the metric placement — only skewness is used.

A **ray** is a line meeting both slits: a transversal of the pair. The set of all rays is the congruence $\mathcal{C}(\ell_1, \ell_2)$. The union $\ell_1 \cup \ell_2$ is the **seam** — the locus that belongs to every ray's addressing but to no ray's interior.

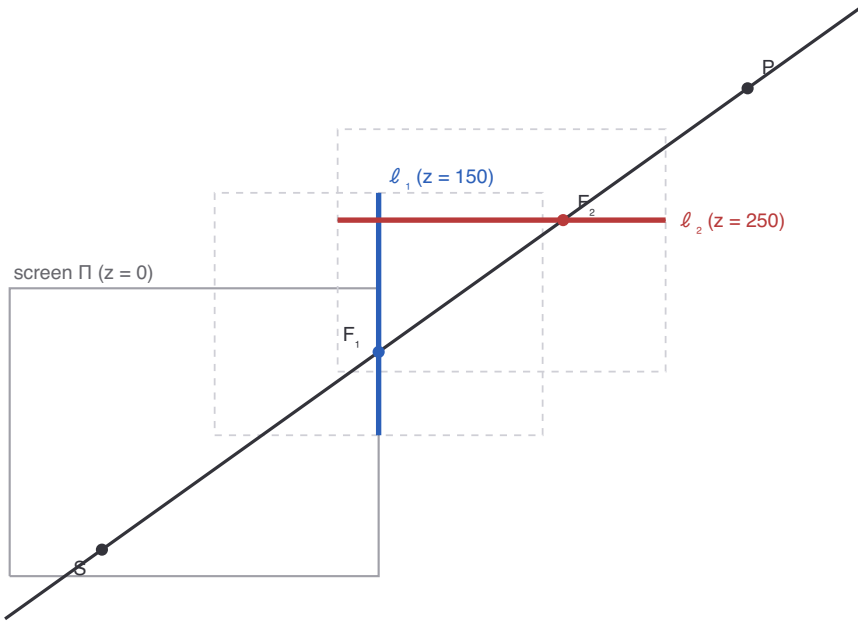


Figure 1. The model, computed in the rig: the screen, the two skew slits, and the unique ray threading the scene point $P = (60, 45, 350)$. Its four canonical points are marked: the feet $F_1 = (0, -45, 150)$ on ℓ_1 and $F_2 = (30, 0, 250)$ on ℓ_2 , and the screen point S . The script asserts their collinearity before drawing them.

PRIMITIVES

D1 — Foot and address. A ray meets ℓ_1 in exactly one point, its first foot $s(r)$, and ℓ_2 in exactly one point, its second foot $t(r)$. The pair (s, t) is the ray's **address**.

D2 — Pencils. Two rays are **1-parallel** if they share the first foot, **2-parallel** if they share the second. The class of all rays through a fixed point $p \in \ell_1$ is the **S-pencil** at p ; likewise T-pencils on ℓ_2 . Each S-pencil is a flat pencil: it fills the plane spanned by p and ℓ_2 . So the S-pencils correspond simultaneously to the points of ℓ_1 and to the planes through ℓ_2 — each slit is read through the other. This is the seam duality.

D3 — Ray space. The address map identifies the set of rays with $\ell_1 \times \ell_2 \cong P^1 \times P^1$. Contrast the pinhole, whose rays through a point form a P^2 . The double slit's ray space is not a projective plane; it is a product of two projective lines — a grid of two interleaved P^1 's. Equivalently, inside the Klein quadric of all lines of P^3 the congruence is itself a smooth

doubly-ruled quadric: the ray space of the doubly-ruled camera is a doubly-ruled surface.

D4 — Chain. A **chain** is the set of rays meeting a fixed line m skew to both slits. Theorem 3 will show a chain is the graph of a Möbius map $\varphi : \ell_1 \rightarrow \ell_2$.

D5 — Screen and image. A screen is a plane containing neither slit. Each ray meets it once, so the screen is a chart of the ray space: image formation is the map *point* $\mapsto$ *its ray* $\mapsto$ *that ray's address*. The image is the ray space itself, read in a chart. In the rig the slits are parallel to the sensor, so the two pencil families become the vertical and horizontal coordinate grid of the screen.



Figure 2. Ray space $P^1 \times P^1$: S-pencils and T-pencils as the two grid families, with one chain drawn in violet — the image of an actual line of the rig, sampled point by point. The script fits a Möbius graph to the samples (residual 7×10^{-14}) and confirms the four marked rays satisfy $\lambda = \mu$ (Theorem 6b) before drawing. Asymptotes dashed.

(*Cross-slit-projection.pdf*, p. 4, right panel: point view of the red slit in top view, picture-plane side view) beside the classical two-point perspective construction. The pairing makes D5 physical: the same drafting moves, with a station point replaced by two slit lines seen edge-on.

IV AXIOMS, GROUP I: THE CAMERA

The first axiom group speaks about rays in space. Its primitives are points, rays, incidence, and the seam.

AXIOMS C1–C3

C1 (Threading). Every point not on the seam lies on exactly one ray.

C2 (Non-concurrence). Two distinct rays have a common point iff they are 1-parallel or 2-parallel; the common point is then the shared foot, on the seam.

C3 (Pencil flatness). The rays of one pencil lie in a common plane through a common seam point; every plane through one slit arises this way from a point of the other.

C1 makes space minus the seam a fiber bundle over the ray space: every point has an address (s, t) plus a position u along its ray — the (S, T, U) chart of the engine, with the seam as singular locus. C2 is the pinhole's negation made law: in a bundle of rays through a point, *every* two rays meet; here two rays meet only by sharing a foot, and the meeting is exiled to the seam. The two axioms are equivalent in the concrete model, but both are stated because they carry the two halves of the intuition — threading is what the camera *does*, non-concurrence is what it *refuses*.

THEOREM 1 — THREADING

In the concrete model, C1–C3 hold. *Proof idea:* the planes spanned by a point P with each slit are distinct and meet in a line through P that meets both slits; any transversal through P lies in both planes, hence equals their intersection. Existence and uniqueness in one stroke.

V AXIOMS, GROUP II: THE IMAGE

Now the rays themselves become the *points* of an abstract plane geometry — the geometry a decoder inhabits, which sees addresses and nothing else. Primitives: a set R of "rays," two equivalence relations $\parallel_1$ and $\parallel_2$, and a family $\mathcal{K}$ of subsets called chains.

AXIOMS M0–M4

M0 (Grid). Every $\parallel_1$ -class meets every $\parallel_2$ -class in exactly one ray.

M1 (Three-Ray). Any three pairwise non-parallel rays lie on exactly one common chain.

M2 (Touching). If K is a chain, $r \in K$, and $q \notin K$ is not parallel to r , there is exactly one chain through q meeting K precisely in r .

M3 (Richness). Each chain meets each parallel class exactly once; four pairwise non-parallel rays exist that share no chain.

M4 (Miquel closure). Given eight rays at the vertices of a combinatorial cube, five of whose faces lie on chains, the sixth face lies on a chain too.

M0–M3 are the axioms of a **Minkowski plane** in the sense of Benz^[3] — the split-signature sibling of the Möbius plane — and M4 is the closure condition that forces the classical model:

THEOREM 2 — REPRESENTATION

A Minkowski plane satisfying M4 is isomorphic to the chain geometry of the projective line over $K \times K$ for a field K ; in the real connected case, to $P^1(\mathbb{R}) \times P^1(\mathbb{R})$ with chains the graphs of Möbius transformations — that is, to the ray space of a double-slit camera. (*Classical; Benz [3], Herzer [7].*)

The algebraic reading deserves its own paragraph. The pinhole's image geometry is coordinatized by a projective plane over $\mathbb{R}$; Möbius geometry is the projective line over $\mathbb{C}$. The double slit's image geometry is the projective line over the **double numbers** $\mathbb{R} \oplus \mathbb{R}$ — pairs added and multiplied componentwise^[6]. Its zero-divisors are the two axes: algebraically, the seam. The number system of double-slit geometry is literally two copies of the real line glued along their arithmetic, with the gluing locus doing the work. In the Benz trinity of circle geometries — $\mathbb{C}$ for Möbius, dual numbers for Laguerre, double numbers for Minkowski — the double slit is the split member.

VI WHAT HAPPENS TO A STRAIGHT LINE

THEOREM 3 — BOWING

The image of a line m of P^3 is: **(1)** a chain — a Möbius graph, in the screen chart a hyperbola with grid-parallel asymptotes — if m is skew to both slits; **(2)** straight exactly when the Möbius denominator degenerates — in the rig, exactly the lines parallel to the screen, plus the pencil lines themselves; **(3)** a **cross** — one full S-pencil with one full T-pencil — if m meets a slit.

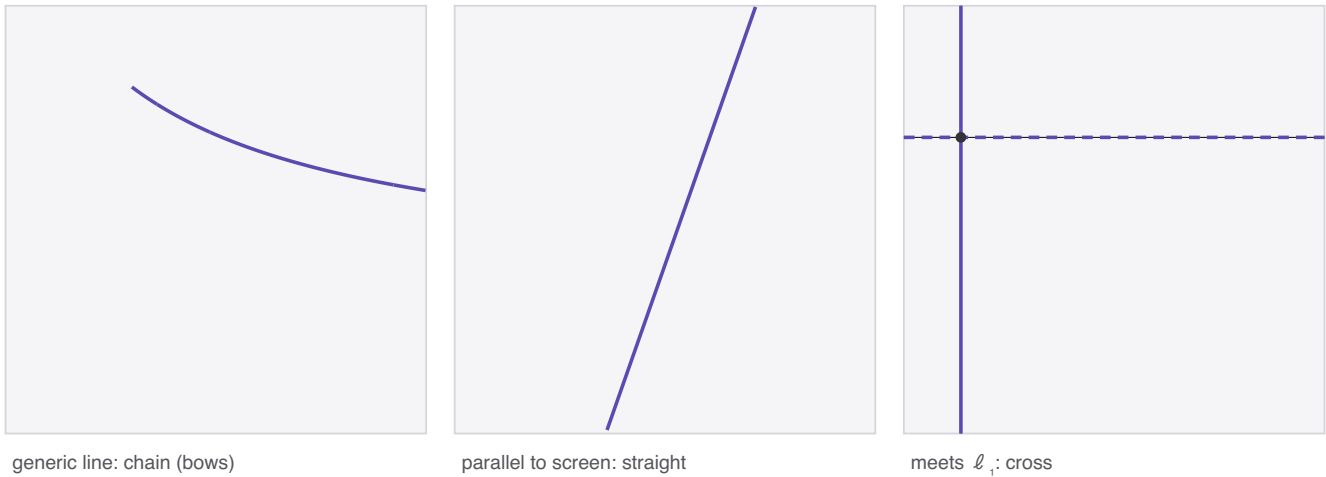


Figure 3. The three cases, each computed by projecting an actual line through the rig. Left: a generic line bows (chord deviation 11 screen units, asserted > 0). Middle: a screen-parallel line stays straight to 3×10^{-14} . Right: a line meeting ℓ_1 images onto a single vertical arm (x-spread 10^{-12}), with the S-pencil at its foot supplying the dashed horizontal arm — the cross.

Straightness, in this geometry, is not generic; it is a coincidence with a name. And the loss of straightness buys something the pinhole cannot have:

THEOREM 4 — REGULUS AMBIGUITY

Two lines skew to the slits have the *same* image iff they belong to a common regulus — one ruling of a quadric whose other ruling contains both slits. The image of a line determines not the line but a one-parameter family; chains correspond bijectively to the quadrics through the slit pair.

grey: rays of one chain (one ruling)
 gold, dashed: every line with the SAME image — the regulus, containing both slits

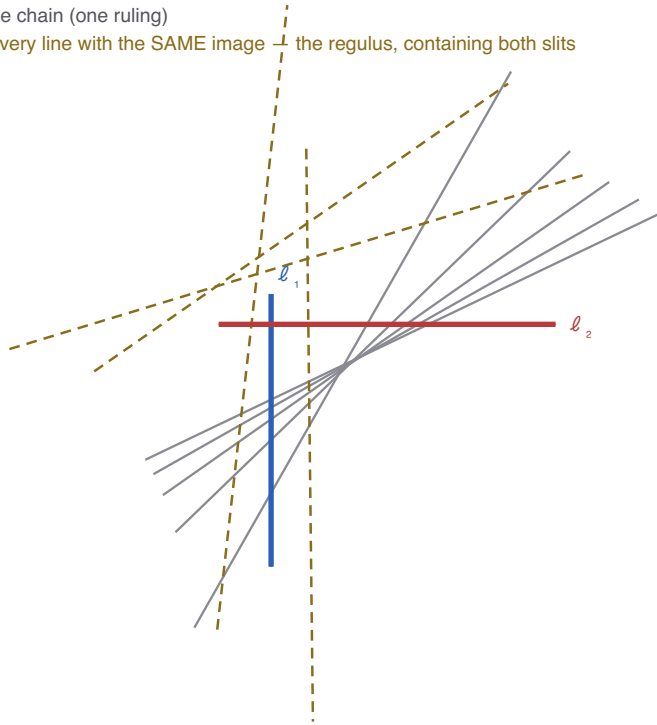


Figure 4. The ambiguity made visible. Grey: rays of one chain (one ruling of the quadric). Gold, dashed: lines with the very same image — the opposite ruling, which contains both slits. The script constructs each gold line from three grey rays and then verifies it meets *all* of them (worst gap 7×10^{-13}), and that the slits meet them too.

Theorem 4 also explains an axiom. **M1**, transported to the concrete model, is the fundamental theorem of projective geometry on P^1 — a homography is determined by three points — wearing the mask of the classical fact that three skew lines determine a quadric. The axiom system knows line geometry without being told.

VII THE GROUP AND THE DEPTH GAUGE

D7. The double-slit group G is the subgroup of $PGL(4, \mathbb{R})$ stabilizing the slit pair.

THEOREM 5 — STRUCTURE (THE ERLANGEN THEOREM)

(1) G has two components — identity and slit-swap — and its identity component has dimension 7, against 15 for the projective group and 12 for the affine. **(2)** G acts on ray space through $(PGL(2, \mathbb{R}) \times PGL(2, \mathbb{R})) \rtimes \mathbb{Z}_2$: independent Möbius actions on the two feet, plus the swap. **(3)** The kernel of that action is a one-parameter **depth gauge**: maps fixing every ray as a set while sliding points along it. **(4)** Conversely, every automorphism of the abstract structure is induced by an element of G .

Part (3) is the group-theoretic root of the oldest fact about images: a single image point carries no depth, because depth is precisely the coordinate the image group cannot see.

filled: a scene · hollow: the gauged scene ($g = 1.6$) — same rays, same image, different depths

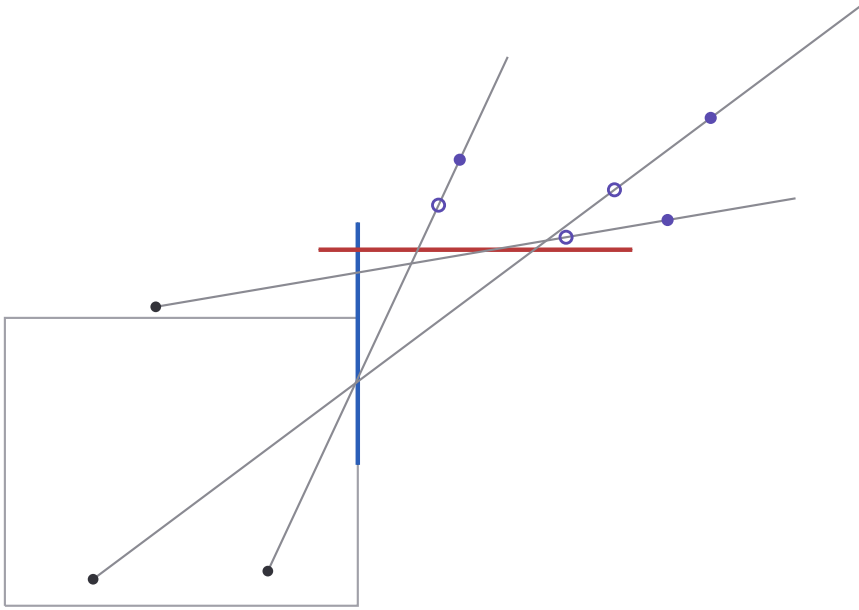


Figure 5. The depth gauge in action, $g = 1.6$. Filled dots: a scene. Hollow dots: the gauged scene — every point slid along its own ray by the Möbius map of depth fixing Z_v and Z_h . The script asserts the addresses are unchanged to 10^{-12} and that the gauge obeys the group law $g \cdot h = gh$. Two different worlds, one image.

VIII THE FUNDAMENTAL INVARIANT

D8. For four pairwise non-parallel rays with addresses (s_i, t_i) , the **bi-cross-ratio** is the pair

$$[[r_1, r_2; r_3, r_4]] = (\lambda, \mu), \lambda = (s_1, s_2; s_3, s_4), \mu = (t_1, t_2; t_3, t_4)$$

— equivalently, one cross-ratio valued in the double numbers.

THEOREM 6A — COMPLETENESS

The bi-cross-ratio is invariant under G 's identity component (the swap exchanges λ and μ), and it is a *complete* invariant: two 4-tuples of pairwise non-parallel rays are group-equivalent

iff their bi-cross-ratios agree up to swap.

THEOREM 6B — CHAIN CRITERION

Four pairwise non-parallel rays lie on a common chain **iff** $\lambda = \mu$ — iff their double-number cross-ratio is real. The discrepancy between λ and μ is a G-invariant measure of a 4-tuple's failure to be the image of one straight line: its **skew defect**.

Where projective geometry attaches one number to four collinear elements, double-slit geometry attaches an ordered pair, and the classical taxonomy of special values doubles: a tetrad can be harmonic on one slit and generic on the other (split-harmonic), harmonic on both, and so on — configurations with no pinhole analogue, because a pinhole has one axis of comparison and a double slit has two that never confer. The reordering group acts diagonally on the pair, and Addendum A9 shows the diagonal action is strict: the classical six-to-three harmonic collapse of the cross-ratio's reordering orbit survives only on the real diagonal — only for co-chained tetrads at harmonic position — while the equianharmonic collapse cannot occur over the double numbers at all. The deepest degeneracy this geometry admits is reserved for chains.

IX THE APERTURE CROSS-RATIO

The theorem that closes the loop between the geometry and the working codec. On the ray through an object point P sit four canonical points: the feet F_1 and F_2 , the screen point S, and P itself.

THEOREM 7 — APERTURE CROSS-RATIO

Let $AR(P)$ be the ratio of the two magnifications of an infinitesimal round target at P as seen on the screen — the codec's measured aspect ratio. Then

$$AR(P) = (F_1, F_2; S, P),$$

the cross-ratio of the four canonical points along P's own ray.

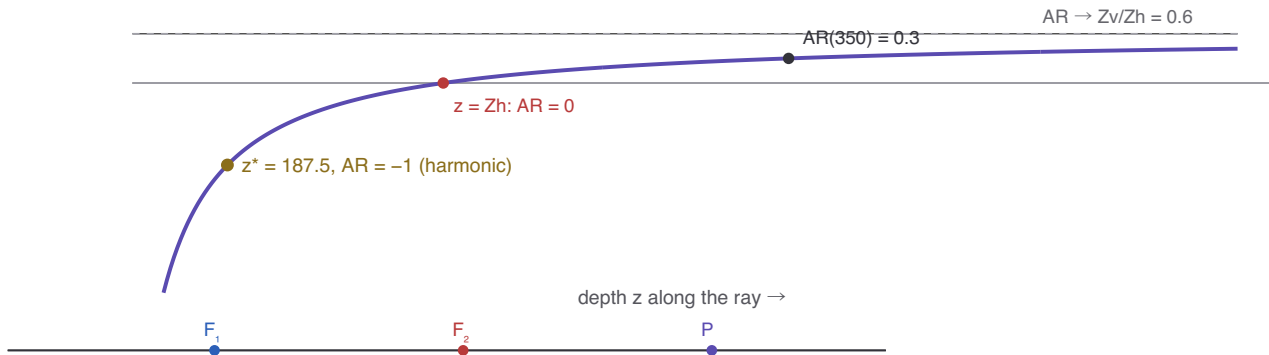


Figure 6. The measured invariant. The curve is $AR(z)$ computed across depth: zero at $z = Zh$, asymptote at $Zv/Zh = 0.6$, and the harmonic point $AR = -1$ at the neutral surface $z^* = 187.5$. Below, the ray as a number line with its four canonical points. The script confirms $AR(350) = 0.3$ equals the cross-ratio to 10^{-12} , and that Solver V2.1 inverts it exactly (round-trip to nine decimals).

Four corollaries, each carrying weight elsewhere in the project. First, **the solver is cross-ratio inversion**: Solver V2.1, $z = Zv \cdot Zh \cdot (AR - 1) / (AR \cdot Zh - Zv)$, is a Möbius function of AR because it inverts a cross-ratio — depth extraction is the oldest projective operation there is, performed along a single ray. Second, **projective invariance**: AR belongs to the point and its four-point configuration, not to the target's extent, which is why size-independence holds. Third, **the Jensen bias is Möbius convexity**: $z(AR)$ is strictly convex on the working interval (asserted in the script), so symmetric noise in AR pushes expected depth systematically outward — the bias measured in the compression gauntlet is a theorem of the geometry, not an artifact of the pipeline. Fourth, **why exactly two slits**: recovering P needs three reference points to coordinatize its ray projectively, and the double-slit camera supplies precisely three — one per slit, one from the screen. One slit gives two points, not enough, and the depth gauge of Theorem 5 remains unfixed; the pinhole collapses $F_1 = F_2$ and loses the coordinate entirely. Two slits is the minimum aperture that makes every ray a measured projective line.

In Klein's manner the geometry refines as elements are nominated and the group shrinks. **Level 1, projective:** data, the slit pair; invariants, incidence, pencils, chains, the bi-cross-ratio, the skew defect — Theorems 1–6 live here. **Level 2, affine:** nominate a plane at infinity; ratios along rays become meaningful, and U can be taken as an affine ratio — the level of the (S, T, U) engine chart. **Level 3, calibrated:** nominate the screen and the metric; the canonical point S joins the feet on every ray and Theorem 7 turns aspect ratio into a coordinate. This is the level of the codec: encoding is evaluating a cross-ratio, transmission is a video stream, decoding is inverting it.

THEOREM 8 — DEGENERATION TABLE

Double-slit geometry contains its neighbors as degenerations of the slit pair: two skew lines give the crossed-slits camera with ray space $P^1 \times P^1$; two lines meeting at a point give the pinhole with ray space P^2 ; one slit at infinity gives the pushbroom; both at infinity give the parallel cameras.

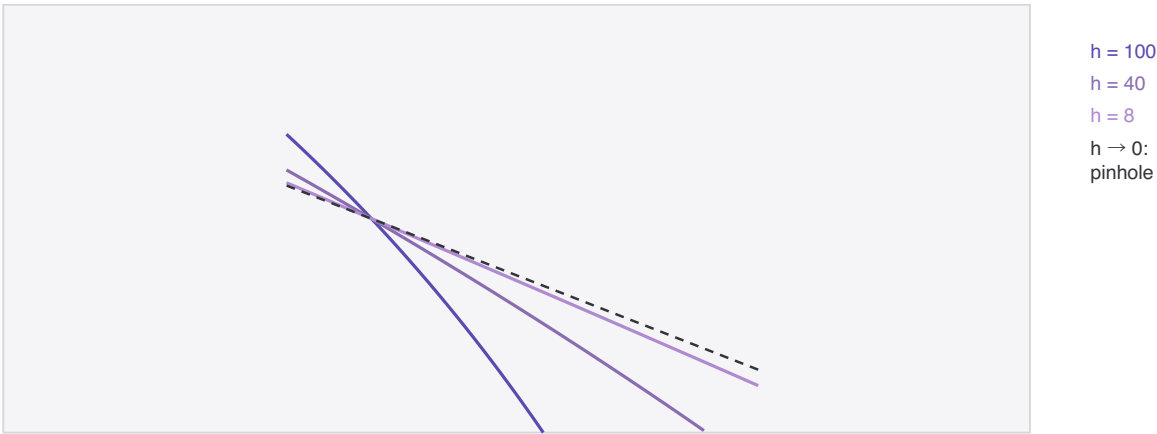


Figure 7. Walking the first row into the second: the image of one fixed scene line as the slit separation h shrinks (violet family), with the pinhole image dashed. The script verifies the mechanism of Addendum A4: the second cross-ratio component μ is *exactly* independent of h (constant to 10^{-12}), and $|\lambda - \mu|$ dies at log–log slope 0.9993 — the fusion is entirely one component walking over to meet the other.

Pinhole / projective geometry	Double-slit geometry
center of projection: a point	slit pair: two skew lines
all rays concur at the center	no two generic rays meet (C2)
ray space P^2	ray space $P^1 \times P^1$, a doubly-ruled quadric

coordinatizing algebra $\mathbb{R} / \mathbb{C}$	double numbers $\mathbb{R} \oplus \mathbb{R}$; seam = zero divisors
image of a line is a line	image of a line is a chain (bows)
line determined by its image	line determined up to a regulus (Thm 4)
cross-ratio λ	bi-cross-ratio (λ, μ) ; chains $\Leftrightarrow \lambda = \mu$
depth lost at the center	depth = cross-ratio against F_1, F_2, S (Thm 7)
Klein ladder: projective $\supset$ affine $\supset$ Euclidean	Levels 1–3 above

One closing remark on the tetrahedron. The engine's root tetrahedron has three pairs of opposite edges: it is not one double-slit geometry but three, sharing a frame, with dynamic axis switching as the passage between them. The atlas of the Tetra Engine is a triple of Minkowski planes stitched along their seams.

Illustration placeholder P3 — the seam in photographs. Insert the "parallel slit planes" axonometric photo collage (*Cross-slit-projection.pdf*, p. 11: emitting image, blue and red slits, ghost image, bridge, receiving image). It belongs here as the one image in the essay where the abstract dictionary above is made of pictures of a horse.

XI EUCLID'S FIVE, REVISITED

The project's founding brief asked for Euclidean geometry with a rewritten fifth postulate. The finished system answers with something better: a precise fate for each of the five, verified like everything else.

E1 — two points determine a line. In the image geometry the count shifts by one: through two non-parallel rays pass *many* chains, and it takes three to determine one (Axiom M1). The script constructs two distinct chains through the same two rays. Lines here behave as circles do in inversive geometry — which is exactly what they are.

E2 — a segment extends indefinitely. It extends, but it returns: chains are closed curves in $P^1 \times P^1$. Every line is a loop through the chart's corner.

E3 — a circle with any center and radius. Circles exist for every center and radius

under the split norm $N(\delta) = \Delta s \cdot \Delta t$ — they are the chains. But the circle of radius zero is not a point: N vanishes on the two pencil directions, so **the radius-zero circle is a cross** — the seam, appearing as the degenerate circle of its own geometry (verified).

E4 — all right angles are congruent. The sleeper. Right angles exist: the perpendicular of a direction of slope m is the direction of slope $-m$, and all right angles are indeed congruent (a boost matches any to any — verified). But the **quarter-turn does not exist**: the script checks both families of split-norm isometries and confirms every one preserves the sign of slope, so no isometry carries a line onto its own perpendicular. The two arms of every right angle lie in different causal sectors, separated by the null directions — and those null directions, the bisectors of every right angle, are the pencil directions. The seam bisects the right angle. What breaks in E4 is not the congruence Euclid asserted but the rotation he assumed beneath it.

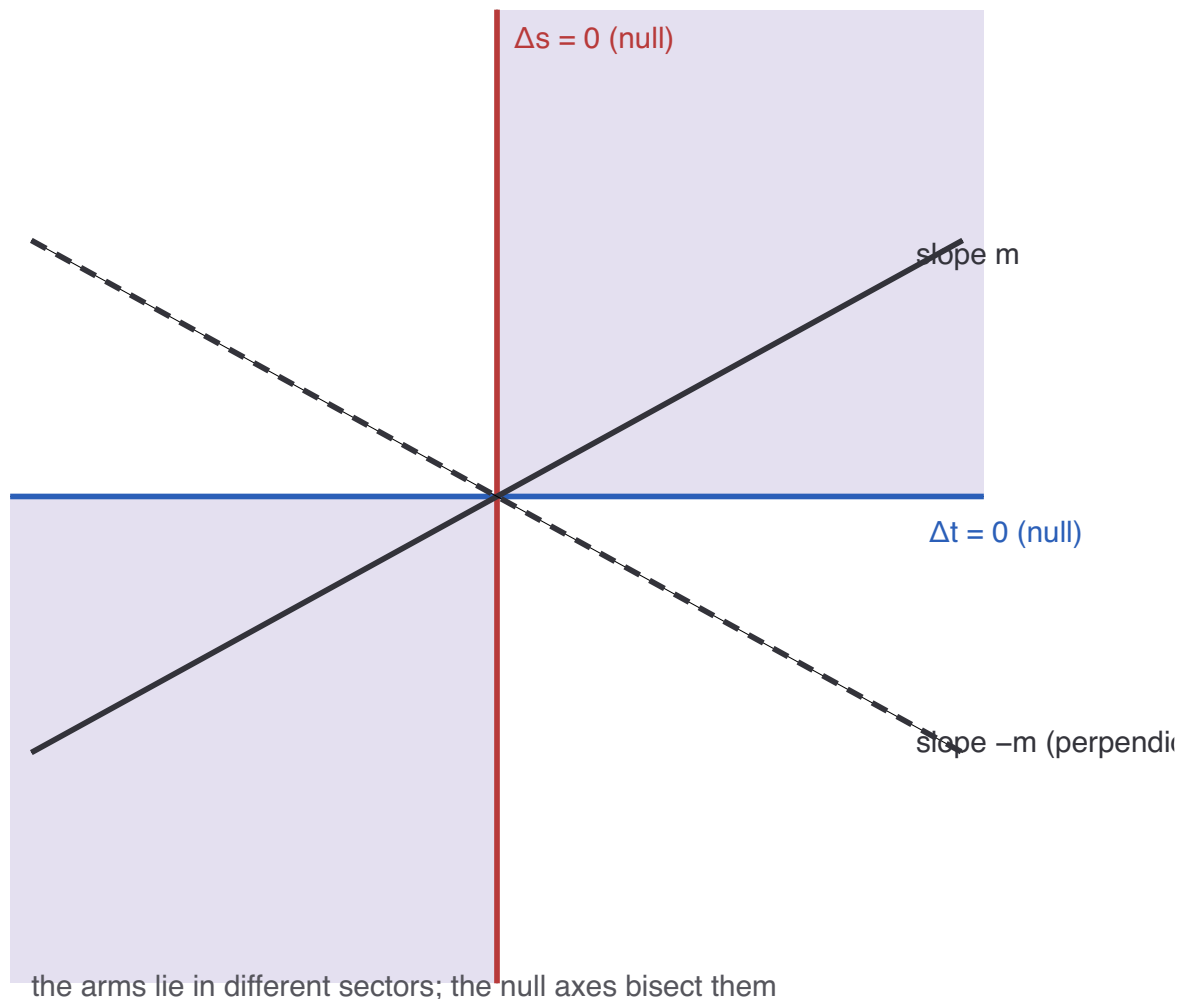


Figure 8. The split tangent plane at a ray. Null directions (the pencil directions, blue and red) divide the directions into sectors; a right angle — slopes m and $-m$ — always straddles them. Both $O(1,1)$ families are verified to preserve the norm and the slope sign: rotation from one arm to the other is not merely absent, it is forbidden.

E5 — the parallel postulate. The one the brief expected to break is the one with the subtlest survival. Parallelism itself *splits in two* ($\parallel_1$ and $\parallel_2$, which never confer), and the structural role of Euclid V — exactly one privileged line through an outside point — is played by the Touching Axiom M2: through a ray off a chain, exactly one chain touching it at a prescribed ray. In double-slit geometry, **parallel survives as tangent**. The postulate the project set out to rewrite was not the one under pressure; the fifth mostly holds where Euclid can see it, and it is the fourth that breaks.

XII REFERENCES

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8. Project documents: [Double-Slit Geometry: Definitions, Axioms, and Theorems](#); [The Seam](#); [Addenda A1–A9](#); [interactive companion](#) and demos 25–34.

